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1) Executive Summary

WannaCry is a ransomware attack that caused a major global cybersecurity incident in May 2017. It combined ransomware capabilities with worm-like propagation, allowing it to spread rapidly across vulnerable Windows systems without requiring direct user interaction. The malware exploited the Windows SMBv1 vulnerability, identified as CVE-2017-0144 (EternalBlue), affecting systems that had not installed Microsoft's security update MS17-010.

Once executed, WannaCry encrypted files on infected machines and demanded ransom payments in Bitcoin in exchange for supposed file recovery. Its rapid propagation affected organizations across multiple sectors, including healthcare, financial services, telecommunications, manufacturing, and transportation. The incident disrupted critical services and demonstrated the serious consequences of unpatched vulnerabilities and inadequate network security.

This threat intelligence investigation examines WannaCry's behavior, attack techniques, indicators of compromise (IOCs), and relevant MITRE ATT&CK techniques. It also reviews threat intelligence platforms such as VirusTotal, AbuseIPDB, AlienVault OTX, MITRE ATT&CK, and URLhaus to understand how security teams can collect and analyze threat-related information.

The investigation highlights the importance of timely security patching, disabling outdated protocols such as SMBv1, network segmentation, endpoint monitoring, and maintaining secure backups. The findings demonstrate how threat intelligence can help organizations identify malicious activity, understand attacker behavior, strengthen defensive controls, and improve their incident response capabilities.

2) Introduction to Threat Intelligence

2.1) what is Cyber Threat Intelligence

Cyber Threat Intelligence (**CTI**) is the process of collecting, processing, analyzing, and interpreting information about cyber threats, threat actors, their motivations, capabilities, tactics, techniques, and procedures (TTPs) to help organizations make informed security decisions and defend against current and future cyberattacks.

2.2) Types of Cyber Threat Intelligence

2.2.1) Strategic Threat Intelligence

Strategic Threat Intelligence is an executive level intelligence focuses on cyber threats, emerging cybersecurity trends, threat actor motivations, and their potential impact on an organization's business operations. It helps senior management make informed security decisions.

Example: Understanding how ransomware attacks are increasing in the healthcare sector and how they could affect patient data, business continuity, and financial operations.

2.2.2) Operational Threat Intelligence

Operation threat intelligence Real-time monitoring of networks, applications and systems to identify vulnerabilities and threats. Analysts and responders use this intelligence to detect and respond to cyberthreats quickly.

Example: Receiving intelligence that a ransomware group is targeting financial institutions through phishing emails and exploiting vulnerable VPN servers.

2.2.3) Tactical Threat Intelligence

Tactical Threat Intelligence focuses on understanding adversaries' Tactics, Techniques, and Procedures (TTPs). It helps security teams understand how attackers operate and improve their detection and prevention mechanisms.

Example: Studying how attackers use phishing, brute-force attacks, privilege escalation, and lateral movement, using the MITRE ATT&CK framework.

2.2.4) Technical Threat Intelligence

Technical Threat Intelligence focuses on technical indicators associated with malicious activities, known as Indicators of Compromise (IOCs). It helps security analysts identify, detect, and investigate potentially malicious activity within an organization's network.

Example: Identifying malicious IP addresses, suspicious domains, malicious file hashes, URLs, and command-and-control (C2) server addresses.

2.3) Indicators of Compromise (IOCs)

An Indicator of Compromise (IOC) is a piece of digital evidence that suggests a system, network, or device may have been compromised or involved in malicious activity. IOCs help security analysts identify, detect, investigate, and respond to potential security incidents.

2.4) Difference between IOC and IOA

IOC	IOA
IOC stands for Indicator of Compromise	IOA stands for Indicator of Attack
Focuses on evidence or artifacts of potential compromise.	Focuses on suspicious activities and attacker behavior.
Can indicate past or ongoing compromise.	Can identify attacks being attempted or carried out.
Example: Known malware file hash	Example: Suspicious PowerShell execution

2.5) Tactics, Techniques and Procedures

- **Tactics**: Explain why an attacker performs an action or what their objective is. For example, gaining initial access or stealing credentials.

- **Techniques**: Explain how an attacker achieves their objective. For example, phishing, brute-force attacks, or exploiting vulnerabilities.

- **Procedures**: Describe the specific steps and tools an attacker uses to execute a technique. For example, using a particular tool to perform lateral movement or privilege escalation.

3) Threat Intelligence tools and Platforms

3.1) VirusTotal

VirusTotal is an online threat intelligence platform used to analyze suspicious files, URLs, domains, and IP addresses. It collects results from multiple antivirus engines and security tools to help analysts identify potentially malicious files and activities.

3.2) AbuseIPDB

AbuseIPDB is a threat intelligence platform used to check IP addresses for reported malicious activities. It helps security analysts identify IP addresses associated with brute-force attacks, spam, port scanning, and other suspicious activities.

3.3) AlienVault OTX

AlienVault Open Threat Exchange (OTX) is an open threat intelligence platform where security researchers and organizations share information about cyber threats. It provides threat indicators such as malicious IP addresses, domains, URLs, and file hashes to help analysts detect potential threats.

3.4) MITRE ATT&CK

MITRE ATT&CK is a globally accessible knowledge base that documents real-world adversary tactics, techniques, and procedures (TTPs). It helps SOC analysts understand attacker behavior, identify attack techniques, develop detection rules, and improve security defenses.

3.5) URLhaus

URLhaus is a threat intelligence platform operated by abuse.ch that collects and shares URLs associated with malware distribution. It helps security analysts identify

malicious URLs, investigate malware-hosting websites, and block dangerous connections.

4) Cyber Threat Investigation

4.1) Selected threat: WannaCry Ransomware

Threat Overview

WannaCry is a ransomware malware campaign that started spreading globally on 12 May 2017. It targeted vulnerable Microsoft Windows systems, encrypted files, and demanded ransom payments in Bitcoin to restore access to affected data. Unlike traditional ransomware, WannaCry contained worm-like capabilities that allowed it to automatically spread across networks by exploiting vulnerable SMBv1 services.

First Incident Date: May 12, 2017

Malware Name: WannaCry also known as **WannaCrypt**

Threat Actor: Lazarus Group

Attackers Objectives

- Financial gain through Bitcoin payments.
- Encrypt organizational files and prevent users from accessing their data.
- Spread malware to additional vulnerable systems.
- Disrupt business operations and critical services.
- Bypassing International Sanctions from North Korea.

4.2) Targeted Organizations

WannaCry was a global campaign that affected organizations and individuals across more than 150 countries.

Major targeted and affected sectors included:

- **Healthcare:** Hospitals and healthcare organizations, particularly the UK's National Health Service (NHS).
- **Financial:** Banks and financial institutions.
- **Telecommunications:** Telecommunications service providers.
- **Manufacturing:** Industrial and manufacturing organizations.
- **Transportation:** Transportation and logistics services.
- **Education:** Universities and educational institutions.

4.3) Initial Access Method and Attack Methodology

WannaCry exploited a vulnerability in Microsoft's Server Message Block (SMBv1) protocol. It used the EternalBlue exploit to target vulnerable Windows systems through CVE-2017-0144, addressed by Microsoft's MS17-010 security update.

Attack Methodology

- **Vulnerability Exploitation:** The malware exploited unpatched Windows systems running vulnerable SMBv1 services over TCP port 445.
- **Remote Code Execution:** EternalBlue allowed malicious code execution on vulnerable systems.
- **Malware Deployment:** The DoublePulsar backdoor mechanism was associated with deploying and executing the WannaCry ransomware payload.
- **Network Propagation:** WannaCry scanned local networks and public IP addresses to identify additional vulnerable systems and spread automatically.
- **File Encryption:** The ransomware encrypted files on infected computers, preventing users from accessing their data.
- **Ransom Demand:** Victims received a ransom message demanding Bitcoin payments in exchange for supposed file recovery.

4.4) Malware tools and technologies involved

WannaCry consisted of four major components that worked together to execute ransomware activities and facilitate communication.

- **DoublePulsar Component:** A backdoor associated with the WannaCry infection mechanism that helped deploy and execute the ransomware payload.
- **Encryption and Decryption Application:** Responsible for encrypting victims' files and providing the mechanism for decrypting them using the required cryptographic keys.
- **Encryption Key Files:** Cryptographic key-related files and data used by the malware to manage file encryption and decryption.
- **Tor Software:** An open-source software component used to facilitate anonymous communication with the ransomware's associated infrastructure.
- **EternalBlue:** Exploit used to target vulnerable SMBv1 services.
- **SMBv1:** Windows file-sharing protocol exploited for network propagation.

4.5) Impact of Attack

WannaCry caused widespread disruption to organizations and individuals around the world, affecting more than 200,000 computers across at least 150 countries.

Major Impacts and Consequences

- **Healthcare Disruption:** The UK's National Health Service (NHS) experienced significant disruption, affecting at least 81 NHS trusts in England.
- **Cancelled Appointments:** Approximately 19,000 NHS appointments were estimated to have been cancelled.
- **Data Inaccessibility:** Infected organizations lost access to important files because of ransomware encryption.
- **Business Disruption:** Hospitals, financial institutions, telecommunications providers, and other organizations experienced operational downtime.
- **Financial Losses:** Organizations faced recovery expenses, IT support costs, lost productivity, and ransom demands.
- **Global Impact:** The campaign demonstrated how a single unpatched vulnerability could enable ransomware to spread across organizations and countries.

4.6) References

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- Kaspersky. (n.d.). *Ransomware WannaCry: All you need to know.* https://www.kaspersky.com/resource-center/threats/ransomware-wannacry

5) Indicators of Compromise (IOC) Investigation

IOC Type	Indicator	Original Source	Intelligence Platform	Why Suspicious	Investigation Date	Detection
Domain	www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com	CISA Alert TA17-132A	Virus Total	Identified by CISA as WannaCry's hard-coded kill-switch domain	11 Nov 2026	12/92 vendors flagged it
File Hash SHA 256	9B60C622546DC45CCA64DF935B71C26DCF4886D6FA811944DBC4E23DB9335640	LogRhythm Labs	Virus Total	Listed in multiple public WannaCry malware-analysis reports as an associated file hash.	11 Nov 2026	68/71 vendors flagged it
IP Address	188.166.23.127	FortinetWannaCry analysis	Abusel PDB, Virus Total	Fortinet identified this IP as a WannaCry-associated C&C server, making it a suspicious network indicator.	11 Nov 2026	3% Abusal PDB, 2% Virus Total
URL	https://nangtamlonto.top/WannaCry.exe	Reported by Amethyst	URL hause	Flaggedon URLhaus as a malware distribution URL directly hosting a Windows executable named WannaCry.exe alongside other malicious payloads.	12 Nov 2026	Listed on URLhaus as malware URL (Tags: WannaCry.exe, Dapato, Status: Offline)
File Name	tasksche.exe	CISA Alert TA17-132A	Virus Total	Malicious executable dropped into C:\WINDOWS\ to execute file encryption	12 Nov 2026	65/69 vendors flagged (SHA-256: ED01EBFB...)
Bitcoin Address	13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94	CISA Alert TA17-132A	Alien Vault OTX	Hard-coded Bitcoin wallet used by WannaCry attackers to collect ransom payments.	12 Nov 2026	Flagged in 9 WannaCry / Lazarus Group threat pulses (BitcoinAddress IOC)
Windows Service	mssecsvc2.0	CISA Alert TA17-132A	Virus Total	Fake Windows service ("Microsoft Security Center (2.0) Service") created by WannaCry to maintain persistence	12 Nov 2026	68/71 vendors flagged it

CVE	CVE-2017-0144	CISA Alert TA17-132A	Alien Vault	Critical SMBv1 remote code execution vulnerability (EternalBlue) exploited by WannaCry to spread across networks.	12 Nov 2026	Indexed as CVE IOC across WannaCry & Lazarus Group threat pulses
Onion Dark Web Address	gx7ekbenv2riucmf.onion	Cisco Talos WannaCry analysis	Virus Total	Hard-coded Tor hidden service domain used by WannaCry for anonymous C2 communication.	12 Nov 2026	3/92 vendors flagged it
Malware tag	WannaCry (tag:WannaCry	CISA Alert TA17-132A	Malware Bazar	Official malware family signature used across threat repositories to classify WannaCry ransomware binaries.	12 Nov 2026	Returned active list of verified WannaCry malware hashes on MalwareBazaar

6) MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Explanation
Execution	Windows Management Instrumentation	T1047	WannaCry executes wmic commands to delete Volume Shadow Copies and inhibit system recovery.
Persistence / Privilege Escalation	Create or Modify System Process: Windows Service	T1543.003	WannaCry creates a fake Windows service named mssecsvc2.0 ("Microsoft Security Center (2.0) Service") to maintain persistence
Defense Evasion	Hidden Files and Directories	T1564.001	WannaCry runs attrib +h to hide its working directory and dropped payload files from users.
Defense Evasion	File and Directory Permissions Modification: Windows File and Directory Permissions Modification	T1222.001	WannaCry uses icacls . /grant Everyone:F /T /C /Q to grant full file access permissions before encrypting data.
Discovery	File and Directory Discovery	T1083	WannaCry scans local drives and folders to locate specific user document, image, and database file extensions for encryption.

Discovery	Peripheral Device Discovery	T1120	WannaCry enumerates attached external storage and removable USB drives to encrypt files stored on them.
Discovery	Remote System Discovery	T1018	WannaCry scans local subnets and random public IP addresses on TCP port 445 to find vulnerable Windows targets.
Discovery	System Network Configuration Discovery	T1016	WannaCry queries local network adapter configurations to identify connected internal networks for worm propagation.
Lateral Movement	Exploitation of Remote Services	T1210	WannaCry exploits the SMBv1 vulnerability (EternalBlue / CVE-2017-0144) to gain remote code execution on unpatched systems
Lateral Movement	Lateral Tool Transfer	T1570	Once a remote machine is exploited via SMB, WannaCry copies and executes its binary payload onto the newly compromised host.

7) Threat Assessment

Question 1: What makes this threat dangerous?

Answer: WannaCry is particularly dangerous because it combines ransomware capabilities with worm-like propagation, allowing it to spread automatically across vulnerable systems and networks without requiring continuous manual intervention from attackers.

Its exploitation of the EternalBlue vulnerability (CVE-2017-0144) in Microsoft's SMBv1 protocol enables remote code execution on unpatched Windows systems. Once a system is infected, WannaCry encrypts files, restricts access to important organizational data, and demands ransom payments in Bitcoin.

Its ability to scan for additional vulnerable systems and spread across networks significantly increases the speed and scale of infection. The attack can disrupt critical services, cause financial losses, interrupt business operations, and make essential information unavailable.

The widespread impact of the 2017 campaign, affecting more than 200,000 computers across at least 150 countries, demonstrates the severity of this threat.

Question 2: Which organizations or industries are most at risk?

Answer: Organizations operating large networks with outdated, unpatched, or unsupported Windows systems are particularly vulnerable to WannaCry attacks.

The industries most at risk include:

- **Healthcare**: Hospitals and healthcare providers that depend on Windows-based systems to manage patient information, appointments, and medical operations.
- **Financial Services**: Banks and financial institutions that rely on interconnected systems and require continuous availability of their services.
- **Telecommunications**: Service providers operating extensive network infrastructures and connected systems.
- **Manufacturing**: Industrial organizations that depend on Windows-based computers for production and operational activities.
- **Transportation**: Transportation and logistics organizations that rely on computer systems to coordinate their operations.
- **Education**: Universities and educational institutions managing large networks of computers and shared resources.

Question 3: What security weaknesses does the threat exploit?

Answer: WannaCry primarily exploits security weaknesses associated with outdated Windows systems and vulnerable SMBv1 network services.

The major weaknesses include:

- **Unpatched Windows Systems**: Systems missing Microsoft's MS17-010 security update remain vulnerable to the EternalBlue exploit.
- **SMBv1 Vulnerability**: The malware exploits CVE-2017-0144 in the SMBv1 protocol to achieve remote code execution.
- **Exposed SMB Services**: Systems allowing unnecessary SMB communication over TCP port 445 may be accessible to attackers.
- **Outdated Operating Systems**: Unsupported or legacy Windows versions may lack essential security updates.
- **Insufficient Network Segmentation**: Poorly segmented networks allow malware to spread between connected systems.
- **Inadequate Security Monitoring**: Insufficient monitoring of suspicious network traffic and malicious activities can delay infection detection.

Question 4: What defensive measures can organizations implement?

Answer: Organizations can implement multiple security measures to prevent WannaCry infections, reduce its ability to spread, and minimize the impact of ransomware attacks.

The recommended defensive measures include:

- **Apply Security Patches**: Install Microsoft's MS17-010 security update to address the SMB vulnerabilities exploited by EternalBlue.

- **Disable SMBv1**: Disable the outdated SMBv1 protocol on systems where it is not required.
- **Network Segmentation**: Divide organizational networks into separate security zones to prevent malware from spreading between connected systems.
- **Endpoint Protection**: Deploy updated antivirus and anti-malware solutions to detect and prevent malicious file execution.
- **Access Control**: Apply the principle of least privilege and restrict unnecessary administrative permissions.
- **Security Monitoring**: Monitor network traffic, endpoint activities, and suspicious SMB connections to identify potential infections.

Question 5: Which IOCs should security teams monitor?

Answer: Security teams should monitor Indicators of Compromise (IOCs) associated with WannaCry to identify potentially infected systems, suspicious network communications, and ransomware-related activities.

The important indicators include:

- **Malicious Domains**: Monitor domains associated with WannaCry, including its hard-coded kill-switch domain.
- **File Hashes**: Monitor known SHA-256 hashes of WannaCry ransomware samples using threat intelligence platforms such as VirusTotal.
- **Malicious IP Addresses**: Investigate suspicious IP addresses associated with reported WannaCry infrastructure or malicious network activities.
- **Malicious URLs**: Monitor URLs associated with WannaCry malware distribution and executable downloads.
- **Malicious File Names**: Look for suspicious executable files such as tasksche.exe in relevant Windows directories.
- **Suspicious Windows Services**: Monitor unexpected service creation, including the reported msseccsvc2.0 service.
- **Network Indicators**: Monitor unusual SMBv1 traffic and suspicious connections over TCP port 445.

Question 6: How could a SOC analyst use this intelligence to detect the threat?

Answer: A Security Operations Center (SOC) team can use the identified Indicators of Compromise (IOCs) to monitor network traffic, investigate suspicious endpoint activity, and detect potential WannaCry infections. By integrating these indicators into security monitoring tools such as SIEM, EDR, and IDS/IPS solutions, analysts can identify suspicious behavior and respond before the ransomware spreads further.

1. **IOC-Based Monitoring**

- Import known malicious file hashes, IP addresses, URLs, and other validated indicators into SIEM and endpoint security platforms. Monitor security logs and network traffic for connections to known malicious infrastructure.
- Use threat intelligence feeds to keep detection rules updated with newly identified indicators.

2. **Network Traffic Analysis**

- Investigate abnormal SMB scanning activity or repeated connection attempts to multiple systems.
- Identify suspicious network behavior that may indicate attempts to exploit vulnerable Windows machines through EternalBlue (CVE-2017-0144).

3. **Endpoint Monitoring**

- Monitor endpoints for suspicious executable files, including known WannaCry-related filenames such as tasksche.exe.
- Investigate unexpected Windows service creation, including suspicious services associated with the malware.
- Detect unusual file modification and encryption activity, which may indicate an active ransomware infection.

4. **Vulnerability Assessment**

- Identify Windows systems that are missing Microsoft's MS17-010 security update.
- Check whether SMBv1 is enabled on organizational systems.
- Prioritize vulnerable machines for immediate patching or isolation to reduce the risk of exploitation.

5. Incident Response

If WannaCry activity is detected, the SOC team should:

- Immediately isolate infected systems from the network to prevent further propagation.
- Block confirmed malicious network indicators where appropriate and restrict unnecessary SMB traffic.
- Preserve relevant system, network, and security logs for forensic investigation.
- Identify other potentially affected machines and assess the scope of the incident.
- Remove WannaCry malware from infected systems and verify that no malicious processes or persistence mechanisms remain.
- Apply the MS17-010 security patch to vulnerable Windows systems and disable SMBv1 wherever it is not required.
- Restore encrypted or affected files from verified clean backups after ensuring the systems are safe.
- Monitor the network and endpoints for signs of reinfection or renewed WannaCry activity.
- Document the incident, its impact, root cause, and lessons learned to improve future security controls.

8) Conclusion

This investigation analyzed the WannaCry ransomware campaign using structured Cyber Threat Intelligence methodologies and passive OSINT platforms including VirusTotal, AbuseIPDB, URLhaus, AlienVault OTX, and MalwareBazaar. By combining technical Indicators of Compromise (IOCs) with behavioral Tactics, Techniques, and Procedures (TTPs), security teams can detect early worm propagation, block malicious infrastructure, and enforce critical patch management (MS17-010) to defend organizational networks against high-impact ransomware threats.